

# Quantum Cryptography: Public Key Distribution and Coin Tossing

Authors: Charles H. Bennett, Gilles Brassard

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## 1. ABSTRACT & THEORETICAL FOUNDATION

We establish the BB84 protocol for unconditionally secure key distribution using non-orthogonal single-photon polarization states, where eavesdropping attempts unavoidably introduce detectable quantum error rates.

## 2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Sending a secret message written with glowing smoke: if a spy tries to peek at it, their gaze immediately alters the smoke, alerting both sender and receiver that they are being watched."

## 3. Mathematical Formulation & Unitary Rule

$$QBER = \frac{\text{Errors}}{\text{Total Sifted Bits}} < 11\%, \quad |\langle 0 | + \rangle|^2$$

Mutually unbiased bases ensure that measuring in the incorrect basis yields purely random outcomes with 50% error probability.

## 4. Step-by-Step Circuit Sequence

Alice applies H or I to encode; Bob applies H or I to measure; classical sifting and parity verification.

## 5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit
import random

# Alice prepares bit 1 in X-basis (| ->)
qc = QuantumCircuit(1, 1)
qc.x(0) # bit = 1
qc.h(0) # basis = X (| ->)

# Bob chooses to measure in X-basis
```

## 6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Vulnerable to Shor's algorithm on ~~quantum networks~~ Information-theoretically secure ~~quantum~~ functional possible future computation

## 7. Key Applications & Future Research Directions

- Government and defense ultra-secure communications
- Banking inter-datacenter encryption links (commercial ID Quantique / Toshiba QKD)
- Satellite-to-ground quantum communications (Micius satellite)

### Future Work:

Decoy state protocols, device-independent QKD, and global satellite-to-ground quantum communications.